Multi-Agent Price Discovery

Analytical HJB vs PPO HJB Market Makers in a Kyle-Style Informed Market

Team

July 16, 2026

Outline

What Are We Trying to Do?

Goal: Study price discovery in a limit-order-book market where:

- An **informed trader** (IT) knows the true fundamental value v and exploits it
- **Market makers** provide liquidity but face adverse selection risk
- We compare three MM configurations: (1) fixed analytical HJB MMs only, (2) learned PPO HJB MMs only, (3) both together
- IT is trained separately in each environment

Key question: Does a learned maker (PPO HJB) behave differently from an analytical one? Does coexistence change price discovery dynamics?

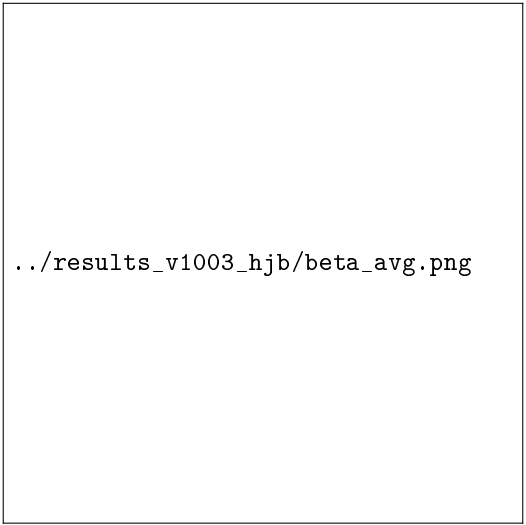
The Kyle-Style Informed Model

Classic Kyle (1985):

- Single informed trader, market makers, noise traders
- IT trades on private information, MMs set prices
- Equilibrium: MM learns λ (price impact), IT chooses order size

Our extension:

- Discrete-time tick-by-tick simulation ($\Delta t = 1$)
- IT action: $\beta \in [0, 10]$ linear policy $q = \beta \cdot (v - a_t)$
- IT reward: $(v - \text{fill price}) \cdot \text{volume}$
- IT is `buy_only` (only exploits positive mispricing)



Market Maker Models

Both MM types use the Guéant HJB framework:

- Solves a PDE backward on inventory grid $q \in [-20, 20]$
- Depth formula: $\delta_{\text{bid}} = \frac{1}{\kappa} + h(q) - h(q+1)$
- κ controls fill-probability decay per tick
- Running penalty ϕ and terminal penalty α on q^2

Analytical HJB MM (fixed)

- $\kappa = 0.8$, constant throughout
- $\lambda = 1.0$, fixed arrival rate
- No learning: quotes deterministically from PDE

PPO HJB MM (trainable)

- Learns $\kappa_t \in [0.01, 2.0]$ via PPO
- 7-dim observation: q, spread, λ est., mid vol, imbalance, t, κ_{t-1}
- Reward: $\Delta\text{cash} - \phi q^2 \Delta t - \alpha q^2 \cdot 1_{\text{last step}}$
- HJB PDE solved once at init with $\kappa_{\text{ref}} = 0.8$

PPO HJB MM: What is Learned vs Estimated

Only κ_t is learned via PPO. The rest of the quoting formula is fixed:

Quoting:

- Deep formula: $\delta_{\text{bid}} = \min\left(\frac{1}{\kappa_t} + h(q) - h(q+1),\ 50\right)$
- $\kappa_t \in [0.01, 2.0]$ — the PPO policy's action
- $h(q)$ comes from HJB PDE solved *once* at init with $\kappa_{\text{ref}} = 0.8$

λ estimation (EMA with 3-tier fallback):

1. **Direct:** If market data provides `md["informed"]`, use it directly:

$$\lambda_{\text{est}} \leftarrow \frac{\texttt{md["informed"]}}{\max(\text{lot_size}, 1)}$$

2. **Taker volume:** Otherwise compute buy-market-order arrival rate:

$$\lambda_{\text{raw}} = \frac{\text{buy_delta}}{\max(\Delta t,\ 10^{-8})}, \qquad \text{buy_delta} = \text{taker_volume_buy}_t - \text{taker_volume_buy}_{t-1}$$

3. **Order-book side changes:** If taker volume unavailable, use cumulated limit-order-book side entries as proxy for buy delta.

EMA smooths everything:

The Market Environment

- Reference price: 1000, tick size: 1, lot size: 1
- Decision interval: $\Delta t = 1.0$, episode length: 40 steps
- Training: 20 000 total timesteps, 40 steps per rollout
- Evaluation: 30 independent episodes (seeds 10 000+ep)
- Rust-backed exchange (`_sim_exchange`), agents interleaved via `OrderShuffler`

Agent	Role	Parameters
Poisson Noise	Random market orders	$\lambda^{\pm} = 1.0$
HJB MM ($\times 3$ or $\times 5$)	Fixed liquidity	$\kappa = 0.8$, $q \in [-20, 20]$
PPO HJB MM ($\times 3$ or $\times 5$)	Learned liquidity	$\kappa \in [0.01, 2.0]$, $q \in [-20, 20]$
Kyle Informed Trader	Directional flow	$\beta \in [0, 10]$, linear, buy-only

Note: Uniform market makers are *removed* from this experiment set to study pure HJB vs PPO dynamics.

Six Experiments

Fair Value v	Configuration	HJB MMs	PPO HJB MMs
$v = 1003$ (3 ticks above ref)	HJB-only	5	0
	PPO-only	0	5
	Mixed (3+3)	3	3
$v = 1006$ (6 ticks above ref)	HJB-only	5	0
	PPO-only	0	5
	Mixed (3+3)	3	3

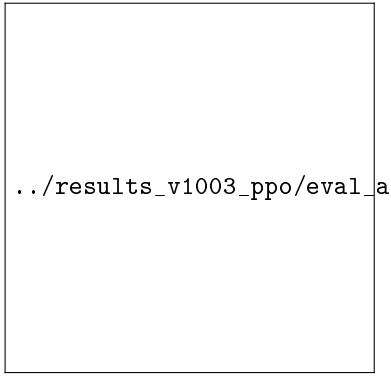
Training: In each experiment, the IT and PPO HJB MMs (if present) train together. **Evaluation:** 30 fresh episodes, all agents in eval mode with trained checkpoints.

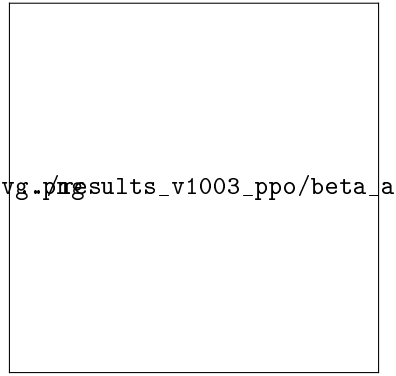
$v = 1003$ — HJB-Only (IT vs 5 analytical HJB MMs)

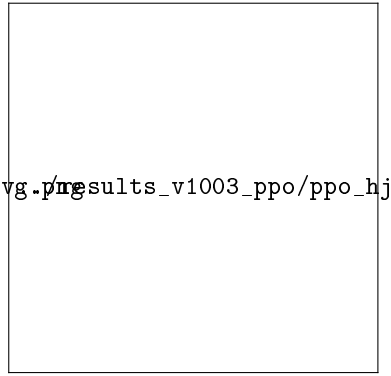




$v = 1003$ — PPO-Only (IT vs 5 PPO HJB MMs)

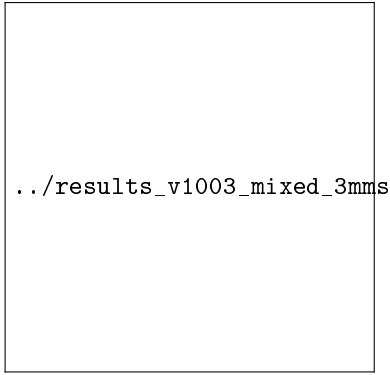


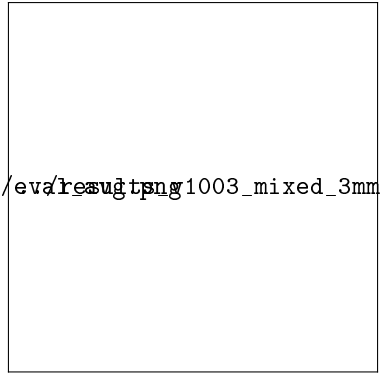


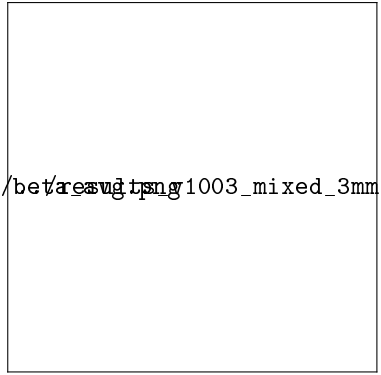


- **Dip-then-rise pattern**: mid drops ~ 997 (below ref) then climbs toward 1003
- κ starts ~ 1.15, dips to ~ 1.07 during mid drop, rises to ~ 1.32 later
- PPO MMs initially widen spreads (lower κ) in response to informed flow, then tighten
- IT's β varies $\sim$ 2–5 depending on observed ask

$v = 1003$ — Mixed (IT vs 3 HJB + 3 PPO HJB MMs)







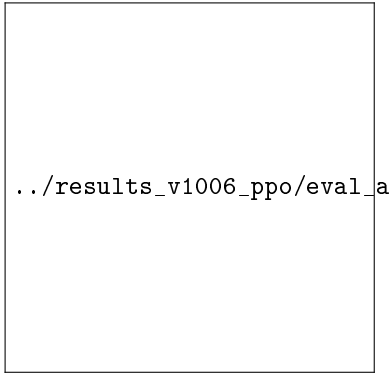
- Smooth convergence toward v — no dip, analytical HJB MMs anchor the mid
- PPO κ converges steadily to ~ 1.25–1.35 over the episode
- IT β oscillates ~ 3–6 — higher than HJB-only (more MM competition = wider spreads)

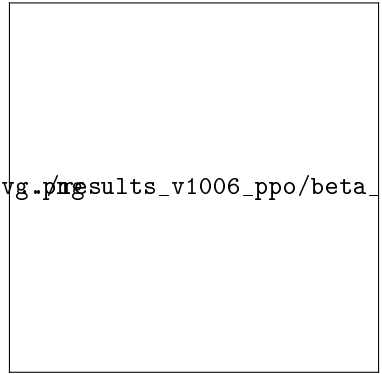
$v = 1006$ — HJB-Only (IT vs 5 analytical HJB MMs)

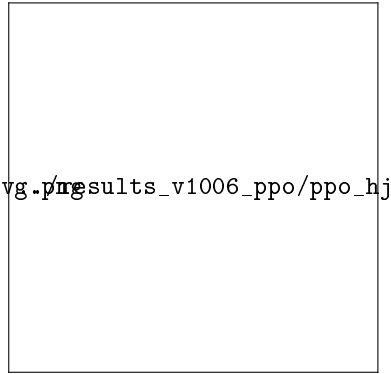




$v = 1006$ — PPO-Only (IT vs 5 PPO HJB MMs)

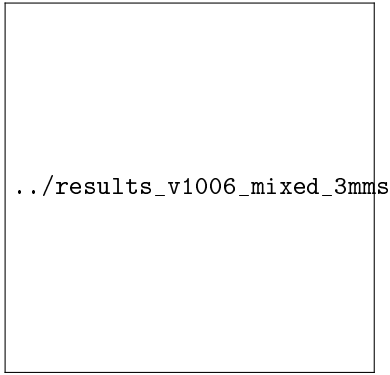


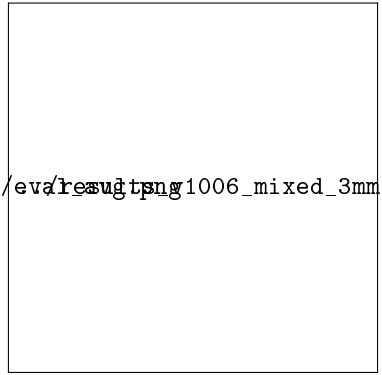


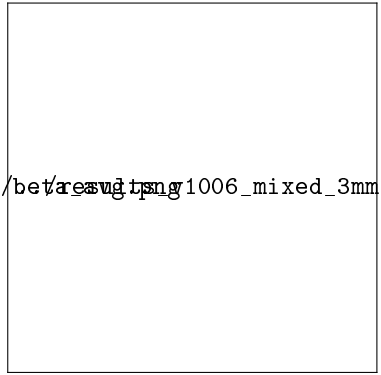


- Dip pattern observed but less pronounced than $v = 1003$
- κ stays high (~ 1.2–1.4) — MMs less reactive because IT demand is already priced in
- Mid ends ~ 1004–1005, about 1–2 ticks below v

$v = 1006$ — Mixed (IT vs 3 HJB + 3 PPO HJB MMs)







- Stable convergence, same pattern as $v = 1003$ mixed
- PPO κ shows wider spread ($\sim$ 1.1–1.3) than $v = 1003$ case — MMs compensate for higher IT aggression
- IT $\beta \sim$ 4–8, the highest across all experiments (more MM targets = more trading opportunity)

HJB vs PPO: Key Differences

PPO-Only Behavior:

- **Dip-then-rise** signature: mid drops below ref_price before converging toward v
- κ drops initially (widening spreads) then rises (tightening)
- MMs learn to *protect* against adverse selection at episode start, then relax
- Without analytical anchor, MMs' collective quoting determines mid fully

Mixed Behavior:

- Smooth convergence — no dip
- Analytical HJB MMs ($\kappa = 0.8$) provide a fixed quoting baseline
- PPO MMs learn to quote slightly tighter ($\kappa > 1.0$) to compete
- IT can exploit both MM types — higher β, more execution

Metric	HJB-Only	PPO-Only	Mixed (3+3)
Mid convergence quality	Steady	Dip-then-rise	Smooth
κ range (PPO MMs)	N/A	1.07–1.32	1.10–1.35
IT β (v=1003)	2–3	2–5	3–6
IT β (v=1006)	4–7	3–7	4–8

Effect of Fair Value v

$v = 1003$ **(3 ticks above ref)**

- Smaller mispricing signal
- IT moderates aggression ($\beta \sim 2$–5)
- Mid converges within ~ 0.5 ticks of v
- Clearer differentiation between MM types

$v = 1006$ **(6 ticks above ref)**

- Larger mispricing $\rightarrow$ more aggressive IT
- IT β reaches 7–8 in mixed case
- Mid ends 1–2 ticks below v — incomplete convergence within 40 steps
- MMs forced wider spreads to survive adverse selection

Takeaway: Larger fair value gaps create stronger adverse selection pressure. The IT captures more surplus; MMs need wider spreads or more steps to fully discover v.

The Dip-Then-Rise: Mechanism

PPO-only experiments show price drops below ref_price (~ 997) before rising toward v (~ 1002.5).

Mechanism:

1. **Step 0:** PPO MMs start with $\kappa \approx 1.15$ (moderate spread). IT begins buying.
2. **Steps 1–10:** MMs observe fills, λ_{est} rises. They **lower** κ **to** ≈ 1.07 (widen spread) as protection.
3. **Asymmetric effect:** IT only hits the ask. No one hits the bid. $\rightarrow$ best bid drifts down, mid = (bid + ask)/2 **drops**.
4. **Steps 10–30:** λ_{est} stabilizes, $h(q)$ inventory penalty already accounts for risk. MMs **raise** κ **to** ≈ 1.32.
5. **Steps 30–40:** Tighter quoting + IT buying drives mid toward v.

Why not in mixed? Analytical HJB MMs ($\kappa = 0.8$, symmetric) keep the bid anchored, preventing the bid from collapsing when PPO MMs widen.

Summary

1. **PPO HJB MMs learn adaptive quoting** — they widen spreads in response to adverse selection and tighten once risk is priced in. This differs fundamentally from the fixed analytical HJB MM.
2. **The dip-then-rise pattern** emerges only in PPO-only environments where no analytical anchor exists. It reveals how learned MMs initially over-protect then re-enter.
3. **Mixed environments** produce the smoothest convergence. Analytical MMs provide a quoting floor; PPO MMs learn to compete by quoting slightly tighter.
4. **Fair value matters**: larger $v - p_0$ gaps amplify adverse selection, increasing IT aggression and slowing price discovery.
5. **3+3 MMs** were sufficient — earlier 5+5 experiments showed instability at $v = 1006$, suggesting too many MMs competing on the same spread disrupts learning.